

Domain Ventilation

code	service	✓ 1						
V-1a	Supply air flow control at the room level	Service group: Air flow control						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No ventilation system or manual control	0	0	0	0	0	0	0
level 1	Clock control	1	0	1	1	1	0	0
level 2	Occupancy detection control	1	0	2	2	2	0	0
level 3	Central Demand Control based on air quality sensors (CO2, VOC, humidity, ...)	2	0	3	3	3	0	0
level 4	Local Demand Control based on air quality sensors (CO2, VOC,...) with local flow from/to the zone regulated by dampers	3	0	3	3	3	0	0
Information sources Standard?								
		EN 15232						

code	service	✗ 0						
V-1c	Air flow or pressure control at the air handler level	Service group: Air flow control						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No automatic control: Continuously supplies of air flow for a maximum load of all rooms	0	0	0	0	0	0	0
level 1	On off time control: Continuously supplies of air flow for a maximum load of all rooms during nominal occupancy time	1	0	0	0	0	0	0
level 2	Multi-stage control: To reduce the auxiliary energy demand of the fan	2	0	0	0	0	0	0
level 3	Automatic flow or pressure control without pressure reset: Load dependent supplies of air flow for the demand of all connected rooms.	3	0	0	0	0	0	0

level 4	Automatic flow or pressure control with pressure reset: Load dependent supplies of air flow for the demand of all connected rooms (for variable air volume systems with VFD).	3	0	0	0	0	0	0
Information sources								
		Standard?	EN 15232					

code	service	✖ 0						
V-2c	Heat recovery control: prevention of overheating	Service group: Air temperature control						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	Without overheating control	0	0	0	0	0	0	0
level 1	Modulate or bypass heat recovery based on sensors in air exhaust	1	0	1	1	1	0	0
level 2	Modulate or bypass heat recovery based on multiple room temperature sensors or predictive control	2	0	2	2	2	0	0
level 3	0	–	–	–	–	–	–	–
level 4	0							
Information sources								
		Standard?	EN 15232					

code	service	✖ 0						
V-2d	Supply air temperature control at the air handling unit level	Service group: Air temperature control						
Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No automatic control	0	0	0	0	0	0	0

level 1	Constant setpoint: A control loop enables to control the supply air temperature, the setpoint is constant and can only be modified by a manual action	1	0	1	1	0	0	0
level 2	Variable set point with outdoor temperature compensation	2	0	2	1	0	0	0
level 3	Variable set point with load dependant compensation. A control loop enables to control the supply air temperature. The setpoint is defined as a function of the loads in the room	3	0	2	1	0	0	0
level 4	0							

Information sources							
	Standard?	EN 15232					

code	service	✗ 0
------	---------	-----

V-3	Free cooling with mechanical ventilation system	Service group: Free cooling
-----	---	-----------------------------

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	No automatic control	0	0	0	0	0	0	0
level 1	Night cooling	1	0	3	2	1	0	0
level 2	Free cooling: air flows modulated during all periods of time to minimize the amount of mechanical cooling	2	0	3	2	1	0	0
level 3	H,x- directed control: The amount of outside air and recirculation air are modulated during all periods of time to minimize the amount of mechanical cooling. Calculation is performed on the basis of temperatures and humidity	3	0	3	2	1	0	0
level 4	0							

Information sources							
	Standard?	EN 15232					

code	service	✓ 1
------	---------	-----

V-6	Reporting information regarding IAQ	Service group: Feedback - Reporting information
-----	-------------------------------------	---

Functionality levels		IMPACTS						
		Energy efficiency	Energy flexibility and storage	Comfort	Convenience	Health, well-being and accessibility	Maintenance and fault prediction	Information to occupants
level 0	None	0	0	0	0	0	0	0
level 1	Air quality sensors (e.g. CO2) and real time autonomous monitoring	0	0	0	0	2	1	1
level 2	Real time monitoring & historical information of IAQ available to occupants	0	0	0	0	3	1	2
level 3	Real time monitoring & historical information of IAQ available to occupants + warning on maintenance needs or occupant actions (e.g. window opening)	0	0	0	0	3	2	3
level 4	0							
Information sources								
Standard?								